

MATSUYAMA, Japan

WMO: 478870

Lat: 33.8435N

Lon: 132.7770E

Elev: 34

StdP: 100.91

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	0.9	-8.7	1.8	4.1	-7.3	2.0	4.7	7.0	5.4	6.3	6.0	2.2	120	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	33.6	25.0	32.7	24.8	31.9	24.6	26.0	31.3	25.5	30.7	25.1	30.1	3.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	19.6	28.6	24.1	19.0	28.3	23.6	18.5	28.2	80.7	31.5	78.7	30.8	76.9	30.3	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.5	-1.6	35.3	1.1	0.9	-2.4	36.0	-3.1	36.5	-3.7	37.0	-4.5	37.6		
(5)			-3.6	26.7	1.3	0.6	-4.5	27.1	-5.3	27.5	-6.0	27.8	-6.9	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	6.1	6.8	9.9	14.9	19.6	23.1	27.5	28.5	25.0	19.5	13.7	8.6	(6)
(7)		DBStd	8.12	2.25	2.77	3.02	3.04	2.18	2.23	2.02	1.68	2.44	2.74	3.04	2.70	(7)
(8)		HDD10.0	324	122	97	39	2	0	0	0	0	0	0	4	60	(8)
(9)		HDD18.3	1546	379	323	262	109	12	0	0	0	0	18	140	302	(9)
(10)		CDD10.0	2876	2	7	36	148	298	393	542	573	449	296	116	17	(10)
(11)		CDD18.3	1054	0	0	0	4	52	143	283	315	199	56	2	0	(11)
(12)		CDH23.3	9500	0	0	1	19	205	797	3016	3684	1562	214	2	0	(12)
(13)		CDH26.7	3404	0	0	0	1	16	166	1175	1572	451	23	0	0	(13)
(14)	Wind	WSAvg	2.1	2.2	2.2	2.3	2.2	2.1	1.8	2.0	2.1	2.0	1.9	2.0	2.2	(14)
(15)	Precipitation	PrecAvg	1486	51	71	110	117	149	240	214	119	160	114	80	61	(15)
(16)		PrecMax	2081	133	127	235	208	299	466	527	347	379	326	163	141	(16)
(17)		PrecMin	806	5	23	45	48	38	31	28	12	58	33	13	4	(17)
(18)		PrecStd	270	28	29	43	41	67	104	129	89	82	76	37	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	18.3	21.1	25.7	28.4	32.4	34.4	34.9	33.1	28.9	23.3	17.7	(19)
(20)			MCWB	10.5	13.2	13.9	17.3	18.7	24.0	25.0	25.0	24.3	22.5	17.5	13.3	(20)
(21)		2%	DB	12.8	15.0	18.8	23.2	26.7	30.0	33.2	33.9	31.6	27.0	21.6	16.0	(21)
(22)			MCWB	9.0	10.3	12.5	15.6	18.0	23.2	24.8	25.1	24.0	20.5	16.4	11.5	(22)
(23)		5%	DB	11.6	13.1	16.9	21.7	25.5	28.5	32.4	33.1	30.5	25.6	20.2	14.6	(23)
(24)			MCWB	7.7	9.0	11.4	14.6	17.4	21.8	24.7	25.0	23.4	19.4	15.5	10.5	(24)
(25)		10%	DB	10.4	11.7	15.2	20.3	24.3	27.2	31.6	32.3	29.3	24.4	18.9	13.4	(25)
(26)			MCWB	6.7	8.0	10.3	13.9	17.0	20.9	24.5	24.8	22.8	18.6	14.7	9.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	14.2	15.6	19.6	22.0	24.7	26.6	26.7	25.7	23.8	19.2	15.2	(27)
(28)			MCDWB	13.9	16.7	18.7	22.9	25.2	29.4	31.9	32.5	30.5	27.0	21.2	16.4	(28)
(29)		2%	WB	10.0	11.6	13.9	17.6	20.5	24.0	25.9	26.1	25.0	22.3	17.9	12.7	(29)
(30)			MCDWB	11.7	13.9	16.9	20.8	23.6	28.5	31.0	31.7	29.4	25.6	20.0	14.7	(30)
(31)		5%	WB	8.5	10.1	12.5	16.2	19.3	23.3	25.4	25.7	24.4	21.0	16.6	11.4	(31)
(32)			MCDWB	10.8	12.4	15.7	19.8	22.8	27.2	30.4	31.1	28.7	24.3	19.4	13.7	(32)
(33)		10%	WB	7.1	8.6	11.3	15.2	18.4	22.5	25.0	25.3	23.9	19.7	15.3	10.0	(33)
(34)			MCDWB	10.0	11.1	14.5	18.9	22.4	25.9	29.9	30.6	27.9	23.0	18.3	12.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	7.4	8.1	8.5	8.4	6.8	6.7	7.1	7.3	7.9	7.7	7.0	(35)
(36)			MCDBR	8.5	9.3	10.2	10.3	10.0	8.3	7.8	8.0	8.3	8.8	8.5	8.6	(36)
(37)			MCWBR	6.0	6.2	6.4	5.3	4.9	3.5	2.8	2.9	3.4	4.4	5.3	5.8	(37)
(38)		5% WB	MCDBR	7.5	7.8	8.5	8.0	7.6	6.7	7.1	7.3	6.9	6.9	7.1	7.2	(38)
(39)			MCWBR	6.2	6.1	6.5	5.1	4.7	3.1	2.8	2.9	3.2	4.1	5.2	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.408	0.454	0.515	0.545	0.546	0.562	0.561	0.532	0.493	0.449	0.422	0.401	(40)	
(41)		taud	2.138	2.036	1.891	1.872	1.934	1.978	2.026	2.111	2.179	2.208	2.194	2.171	(41)	
(42)		Ebn at Noon	773	777	760	760	764	749	747	763	774	775	755	754	(42)	
(43)		Edh at Noon	122	150	188	200	190	182	172	156	140	125	115	112	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.38	3.02	4.05	4.87	5.28	4.51	5.03	5.17	4.15	3.43	2.56	2.07	(44)	
(45)		RadStd	0.19	0.24	0.30	0.47	0.54	0.52	0.65	0.51	0.41	0.34	0.22	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page